

④ j is odd, k is odd. By Defn. 1, $\exists$ integers K_1 and K_2 s.t. $j = 2K_1 + 1$ and $k = 2K_2 + 1$. $\therefore (j-k) = (2K_1+1) - (2K_2+1) = 2(K_1-K_2)$

By Defn. 1, $(j-k)$ is an ~~odd~~ even integer.

$|j-k| = j-k$ which is even, if $j-k \geq 0$

Else, $|j-k| = -(j-k) = -2(K_1-K_2) = 2(K_2-K_1)$

By Defn. 1, $|j-k|$ is even. $\square$

Let p be an odd integer. By Defn. 1, $\exists$ integer a s.t. $p = (2a+1)$

$\therefore p-1 = (2a+1)-1 = 2a$. By Defn. 1, $p-1$ is even

$\therefore p+1 = (2a+1)+1 = 2(a+1)$. By Defn. 1, $p+1$ is even

$\therefore p-2 = (2a+1)-2 = 2a-1 = 2(a-1)+1$. By Defn. 1, $p-2$ is odd

Let p be an even integer. By Defn. 1, $\exists$ integer a s.t. $p = 2a$

$\therefore p+1 = (2a+1)$. By Defn. 1, $p+1$ is odd

$\therefore p-1 = 2a-1 = 2(a-1)+1$. By Defn. 1, $p-1$ is odd

$\therefore p-2 = 2a-2 = 2(a-1)$. By Defn. 1, $p-2$ is even.

~~Conclusion:~~ Conclusion: p has opposite parity with $p-1$ and $p+1$, but

has the same parity with $(p-2)$.

Let, before a given step in the process, there be x even integers and y odd integers.

On execution of Case 1, there are $(x-2)+1 = (x-1)$ even, y odd integers.
 $\therefore$ The parity of even integers has changed, but odd integers have the same parity.

On execution of Case 2, there are $(x-1)+1 = x$ even, $(y-1)+1 = y$ odd integers.
 $\therefore$ Even integers $\rightarrow$ Parity changed Odd integers $\rightarrow$ same parity

On execution of Case 3, there are $(x-1)+1 = x$ even, $(y-1)+1 = y$ odd integers.
 $\therefore$ Even integers $\rightarrow$ Parity changed Odd integers $\rightarrow$ same parity